

Sequentialization for Multiplicative Proof-Nets

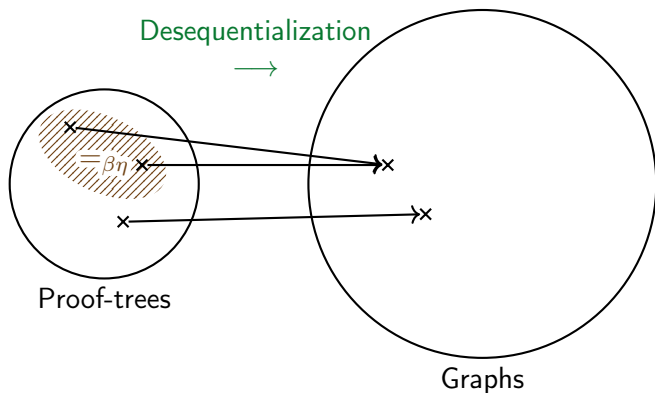
Rémi Di Guardia

1st October 2026



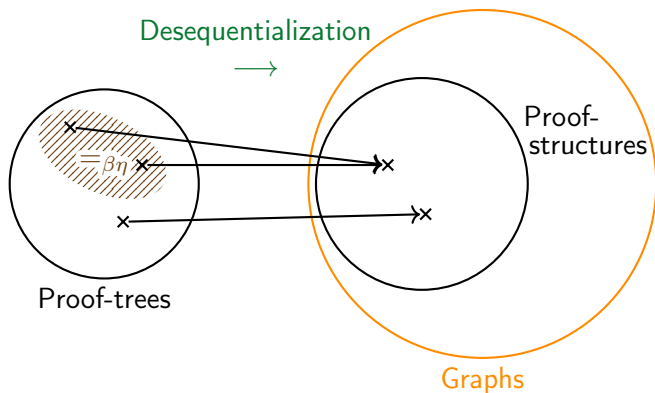
From Proof-Nets back to Proof-Trees

Proof-Nets graphical syntax for proofs of **Linear Logic**, *canonical*



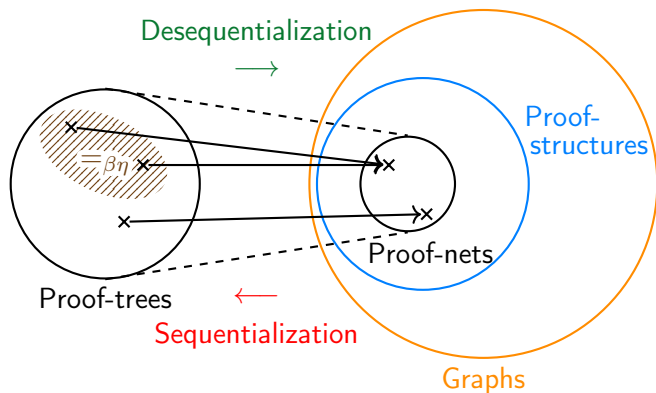
From Proof-Nets back to Proof-Trees

Proof-Nets graphical syntax for proofs of **Linear Logic**, *canonical*



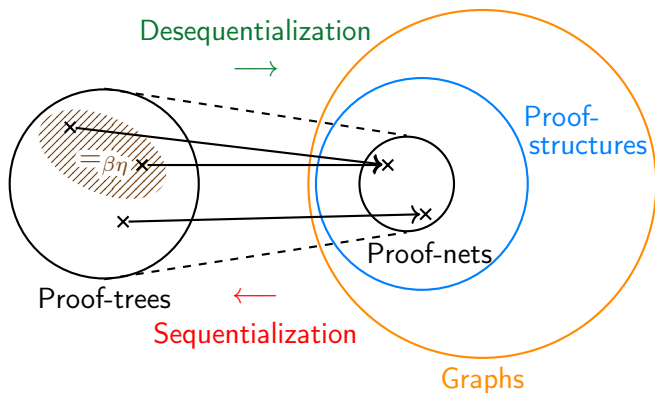
From Proof-Nets back to Proof-Trees

Proof-Nets graphical syntax for proofs of **Linear Logic**, *canonical*



From Proof-Nets back to Proof-Trees

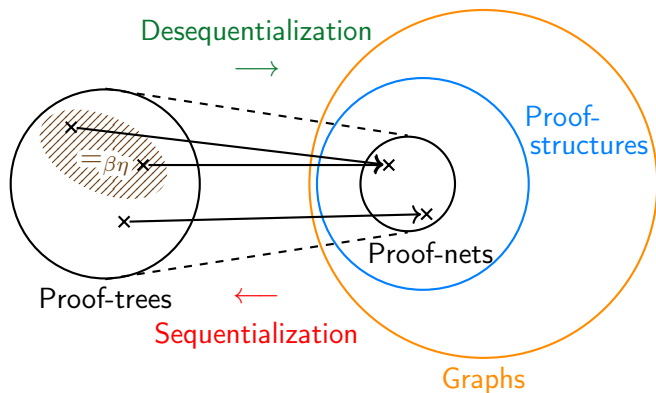
Proof-Nets graphical syntax for proofs of **Linear Logic**, *canonical*



Multiple **correctness criteria**, proofs of sequentialization

From Proof-Nets back to Proof-Trees

Proof-Nets graphical syntax for proofs of **Linear Logic**, *canonical*

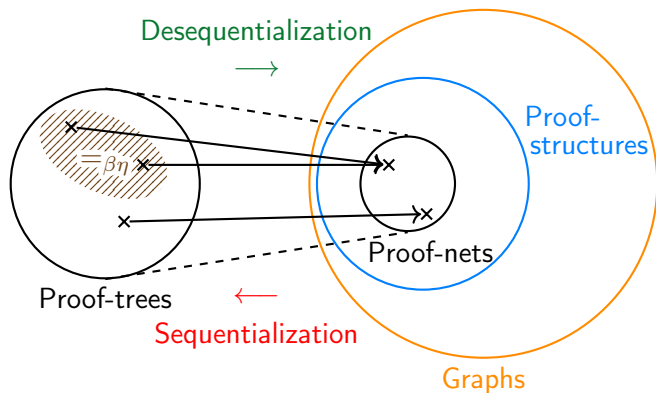


Multiple **correctness criteria**, proofs of sequentialization

Today an “easy” proof of sequentialization

Outline

- ▶ Reminder: Deserialization & Proof-Nets
- ▶ Proof of Sequentialization



Unit-free Multiplicative Linear Logic

Formulas

$$A ::= X \mid X^\perp \mid A \otimes A \mid A \wp A$$

Negation

$$(X^\perp)^\perp = X \quad (A \otimes B)^\perp = A^\perp \wp B^\perp \quad (A \wp B)^\perp = A^\perp \otimes B^\perp$$

Sequents

$$\vdash A_1, A_2, \dots, A_n$$

Rules

$$\frac{}{\vdash A^\perp, A} \text{ (ax)} \quad \frac{\vdash A^\perp, \Gamma \quad \vdash A, \Delta}{\vdash \Gamma, \Delta} \text{ (cut)}$$

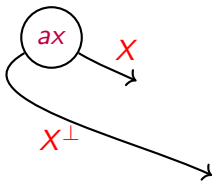
$$\frac{\vdash A, \Gamma \quad \vdash B, \Delta}{\vdash A \otimes B, \Gamma, \Delta} (\otimes) \quad \frac{\vdash A, B, \Gamma}{\vdash A \wp B, \Gamma} (\wp)$$

Desequentialization: From Trees to Graphs

$$\begin{array}{c}
 \frac{}{\vdash X^\perp, X} \text{ (ax)} \quad \frac{}{\vdash Y, Y^\perp} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \quad \frac{}{\vdash Z^\perp, Z} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \otimes Z^\perp, Z \quad \text{ (}\otimes\text{)} \\
 \hline
 \vdash X \otimes Y, X^\perp \wp (Y^\perp \otimes Z^\perp), Z \quad \text{ (}\wp\text{)} \\
 \hline
 \vdash X \otimes Y, (X^\perp \wp (Y^\perp \otimes Z^\perp)) \wp Z \quad \text{ (}\wp\text{)}
 \end{array}$$

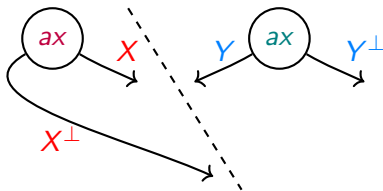
Desequentialization: From Trees to Graphs

$$\begin{array}{c}
 \frac{}{\vdash X^\perp, X} \text{ (ax)} \quad \frac{}{\vdash Y, Y^\perp} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \quad \frac{}{\vdash Z^\perp, Z} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \otimes Z^\perp, Z \quad \text{ (}\otimes\text{)} \\
 \hline
 \vdash X \otimes Y, X^\perp \wp (Y^\perp \otimes Z^\perp), Z \quad \text{ (}\wp\text{)} \\
 \hline
 \vdash X \otimes Y, (X^\perp \wp (Y^\perp \otimes Z^\perp)) \wp Z \quad \text{ (}\wp\text{)}
 \end{array}$$



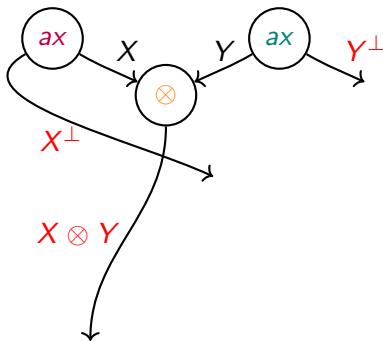
Desequentialization: From Trees to Graphs

$$\begin{array}{c}
 \frac{}{\vdash X^\perp, X} \text{ (ax)} \quad \frac{}{\vdash Y, Y^\perp} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \quad \frac{}{\vdash Z^\perp, Z} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \otimes Z^\perp, Z \quad \text{ (}\otimes\text{)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \otimes Z^\perp, Z \quad \text{ (}\wp\text{)} \\
 \hline
 \vdash X \otimes Y, X^\perp \wp (Y^\perp \otimes Z^\perp), Z \quad \text{ (}\wp\text{)} \\
 \hline
 \vdash X \otimes Y, (X^\perp \wp (Y^\perp \otimes Z^\perp)) \wp Z \quad \text{ (}\wp\text{)}
 \end{array}$$



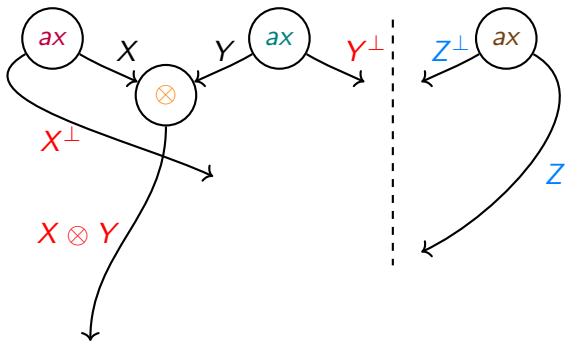
Desequentialization: From Trees to Graphs

$$\begin{array}{c}
 \frac{}{\vdash X^\perp, X} \text{ (ax)} \quad \frac{}{\vdash Y, Y^\perp} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \quad \frac{}{\vdash Z^\perp, Z} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \otimes Z^\perp, Z \quad (\otimes) \\
 \hline
 \vdash X \otimes Y, X^\perp \wp (Y^\perp \otimes Z^\perp), Z \quad (\wp) \\
 \hline
 \vdash X \otimes Y, (X^\perp \wp (Y^\perp \otimes Z^\perp)) \wp Z \quad (\wp)
 \end{array}$$



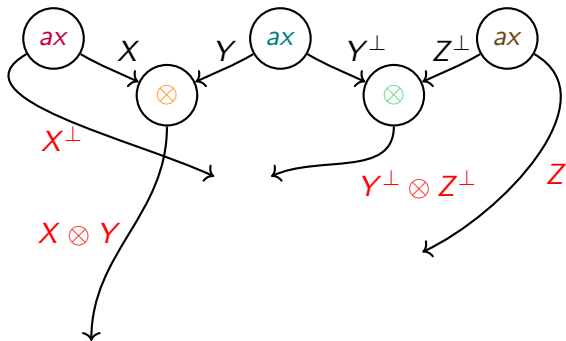
Desequentialization: From Trees to Graphs

$$\begin{array}{c}
 \frac{}{\vdash X^\perp, X} \text{ (ax)} \quad \frac{}{\vdash Y, Y^\perp} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \quad (\otimes) \quad \frac{}{\vdash Z^\perp, Z} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \otimes Z^\perp, Z \quad (\otimes) \\
 \hline
 \vdash X \otimes Y, X^\perp \wp (Y^\perp \otimes Z^\perp), Z \quad (\wp) \\
 \hline
 \vdash X \otimes Y, (X^\perp \wp (Y^\perp \otimes Z^\perp)) \wp Z \quad (\wp)
 \end{array}$$



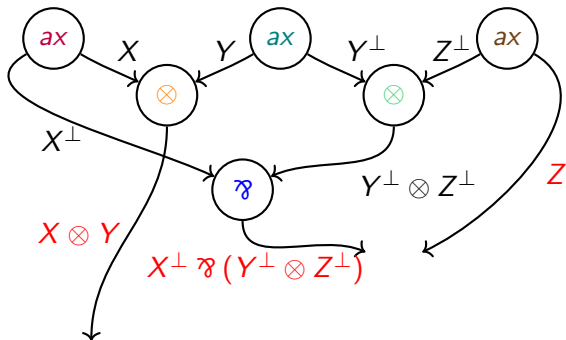
Desequentialization: From Trees to Graphs

$$\begin{array}{c}
 \frac{}{\vdash X^\perp, X} \text{ (ax)} \quad \frac{}{\vdash Y, Y^\perp} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \quad (\otimes) \quad \frac{}{\vdash Z^\perp, Z} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \otimes Z^\perp, Z \quad (\otimes) \\
 \hline
 \vdash X \otimes Y, X^\perp \wp (Y^\perp \otimes Z^\perp), Z \quad (\wp) \\
 \hline
 \vdash X \otimes Y, (X^\perp \wp (Y^\perp \otimes Z^\perp)) \wp Z \quad (\wp)
 \end{array}$$



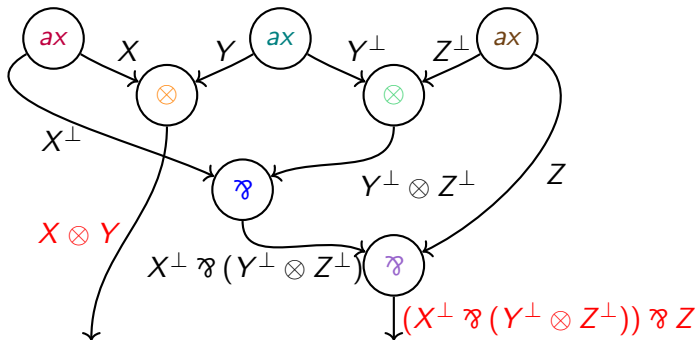
Desequentialization: From Trees to Graphs

$$\begin{array}{c}
 \frac{}{\vdash X^\perp, X} \text{ (ax)} \quad \frac{}{\vdash Y, Y^\perp} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \quad \text{ (}\otimes\text{)} \quad \frac{}{\vdash Z^\perp, Z} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \otimes Z^\perp, Z \quad \text{ (}\otimes\text{)} \\
 \hline
 \vdash X \otimes Y, X^\perp \wp (Y^\perp \otimes Z^\perp), Z \quad \text{ (}\wp\text{)} \\
 \hline
 \vdash X \otimes Y, (X^\perp \wp (Y^\perp \otimes Z^\perp)) \wp Z \quad \text{ (}\wp\text{)}
 \end{array}$$



Desequentialization: From Trees to Graphs

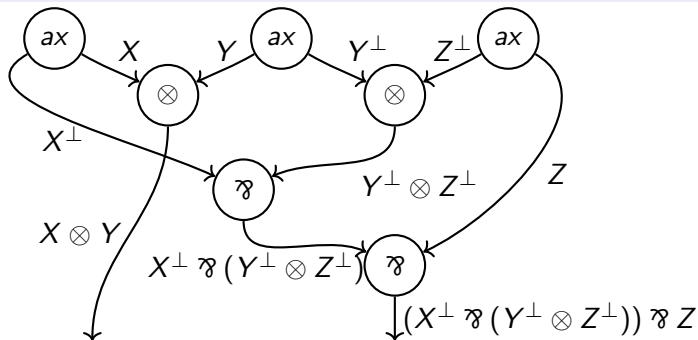
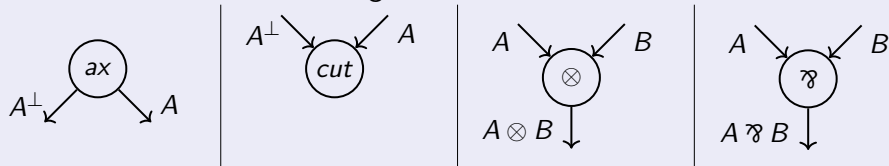
$$\begin{array}{c}
 \frac{}{\vdash X^\perp, X} \text{ (ax)} \quad \frac{}{\vdash Y, Y^\perp} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \quad \frac{}{\vdash Z^\perp, Z} \text{ (ax)} \\
 \hline
 \vdash X \otimes Y, X^\perp, Y^\perp \otimes Z^\perp, Z \quad \text{ (}\otimes\text{)} \\
 \hline
 \vdash X \otimes Y, X^\perp \wp (Y^\perp \otimes Z^\perp), Z \quad \text{ (}\wp\text{)} \\
 \hline
 \vdash X \otimes Y, (X^\perp \wp (Y^\perp \otimes Z^\perp)) \wp Z \quad \text{ (}\wp\text{)}
 \end{array}$$



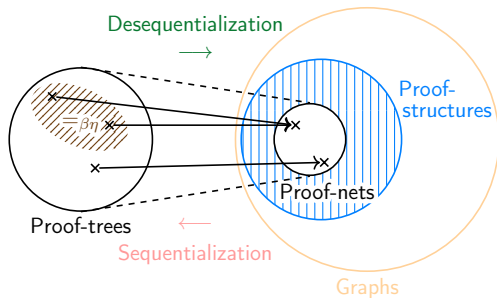
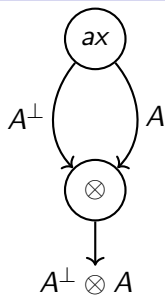
Proof-Structure

Definition: Proof-Structure

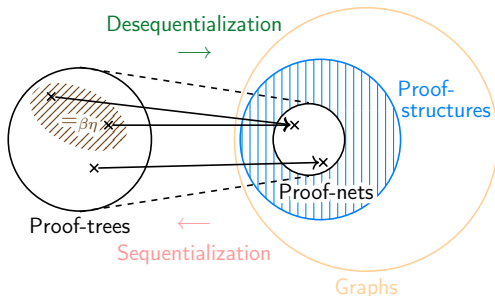
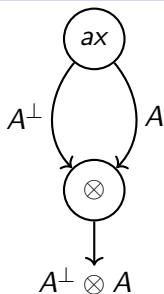
Partial graph with labels on vertices $\rightarrow ax, cut, \otimes, \wp$
 on edges $\rightarrow$ formula



Correctness Criterion

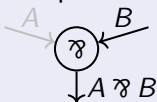


Correctness Criterion



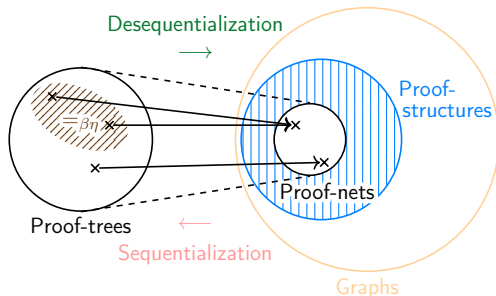
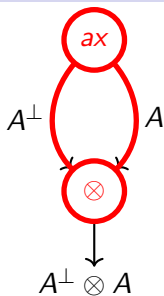
Definition: Danos-Regnier Correctness Criterion

Correctness graph: remove one premise of each $\wp$



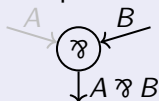
Correct when all correctness graphs are *acyclic* and *connected* (non-oriented paths)

Correctness Criterion



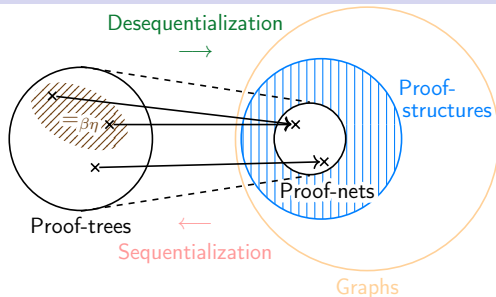
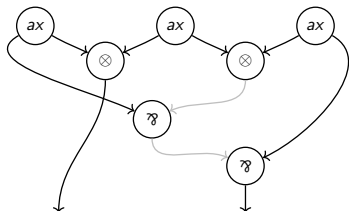
Definition: Danos-Regnier Correctness Criterion

Correctness graph: remove one premise of each $\wp$



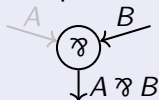
Correct when all correctness graphs are *acyclic* and *connected* (non-oriented paths)

Correctness Criterion



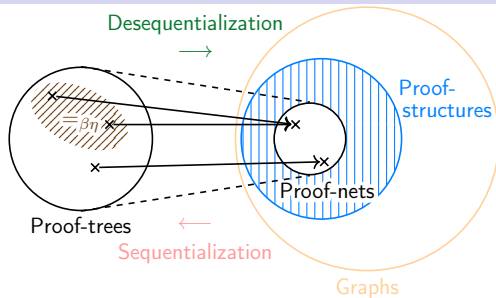
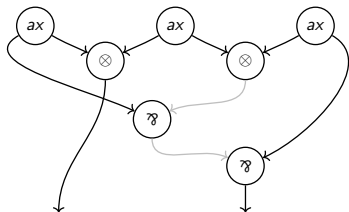
Definition: Danos-Regnier Correctness Criterion

Correctness graph: remove one premise of each $\wp$



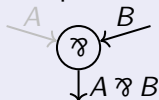
Correct when all correctness graphs are *acyclic* and *connected* (non-oriented paths)

Correctness Criterion



Definition: Danos-Regnier Correctness Criterion

Correctness graph: remove one premise of each $\wp$



Correct when all correctness graphs are **acyclic** and **connected** (non-oriented paths)

Switching path: does not contain the two premises of a $\wp$

$\leadsto$ **No switching cycle**

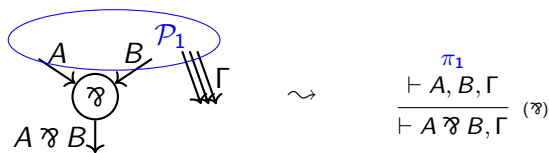
Sequentialization

Theorem: Sequentialization

Given a correct proof-structure, there is a proof-tree desequentializing to it.

How to prove it? Find a vertex corresponding to a **last rule**

- a $\wp$ with no vertex below



- a $\otimes$ with no vertex below and not in a cycle



- a *cut* not in a cycle
 $\longrightarrow$ same as a $\otimes$, we will not consider *cut*

Building a Switching Path

Idea of the proof build a *path* p by following a chain of **dependencies**
 $\leadsto$ before doing this $\otimes$ -rule we must first do this $\wp$ -rule, before which we
must do this $\otimes$ -rule ... stopping on a rule we can actually do
(c.f. previous drawings)

Building a Switching Path

Idea of the proof build a *path* p by following a chain of **dependencies**
 $\leadsto$ before doing this $\otimes$ -rule we must first do this $\wp$ -rule, before which we
must do this $\otimes$ -rule ... stopping on a rule we can actually do
(c.f. previous drawings)

Plan

- 1 We build a **switching path** p by visiting vertices, halting our construction only on a vertex corresponding to a last rule.
- 2 We prove p **cannot “loop”**: its vertices are distinct.

Then we are done!

- The graph is finite $\longrightarrow$ the path's construction must stop, necessarily on a vertex corresponding to a last rule!
- We remove this vertex and conclude by induction hypothesis.

Building a Switching Path

Idea of the proof build a *path* p by following a chain of **dependencies**
 $\leadsto$ before doing this $\otimes$ -rule we must first do this $\wp$ -rule, before which we
must do this $\otimes$ -rule ... stopping on a rule we can actually do
(c.f. previous drawings)

Plan

- 1 We build a **switching path** p by visiting vertices, halting our construction only on a vertex corresponding to a last rule.
- 2 We prove p **cannot “loop”**: its vertices are distinct.

Then we are done!

- The graph is finite $\longrightarrow$ the path's construction must stop, necessarily on a vertex corresponding to a last rule!
- We remove this vertex and conclude by induction hypothesis.

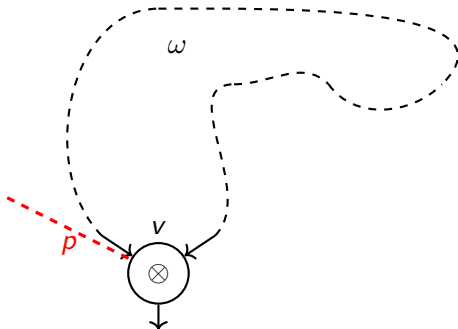
Initial p a premise of any arbitrary $\otimes$ or $\wp$ vertex v with no vertex below it
(if only ax vertices, then easy to conclude)

Ascending Path α : a $\otimes$ blocked by a $\wp$

Assume p ends with a premise of a vertex v with no one below it.

- If v is a $\wp$ or is a $\otimes$ not in a cycle: we are done.
- Otherwise: v is a $\otimes$ in a cycle ω .

Remember ω cannot be switching: it has at least one  ;
w.l.o.g. ω has a minimal number of those (used later to not loop).

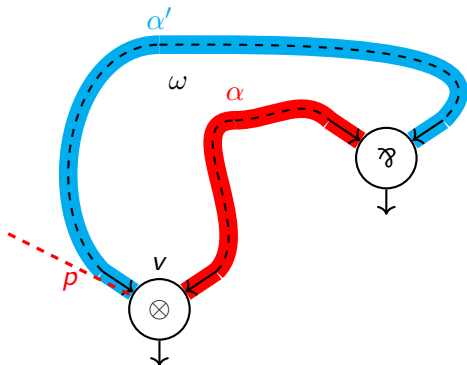


Ascending Path α : a $\otimes$ blocked by a $\wp$

Assume p ends with a premise of a vertex v with no one below it.

- If v is a $\wp$ or is a $\otimes$ not in a cycle: we are done.
- Otherwise: v is a $\otimes$ in a cycle ω .

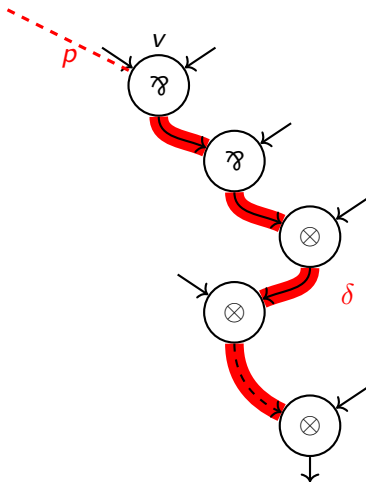
Remember ω cannot be switching: it has at least one  ;
w.l.o.g. ω has a minimal number of those (used later to not loop).
Extend p with the sub-path α of ω towards the first .



Descending Path δ : formula dependency

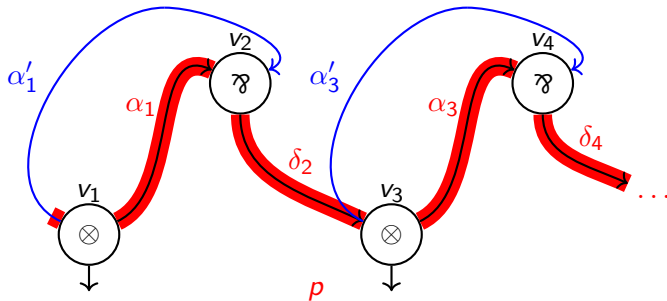
Assume p ends with a premise of a $\mathcal{T}$ -vertex v with someone below it.

Extend p with **downward edges δ** towards a vertex with no one below it.



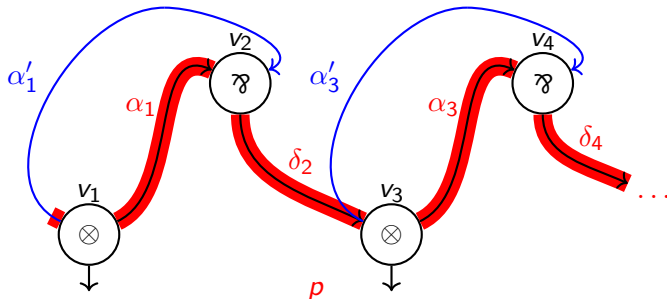
Built Switching Path p

Concatenation of an ascending path, a descending path, ...



Built Switching Path p

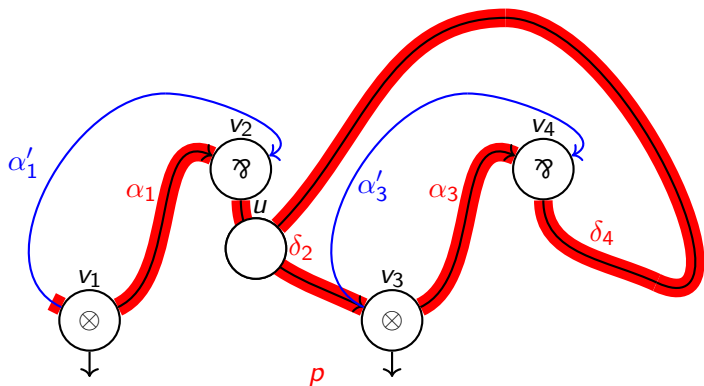
Concatenation of an ascending path, a descending path, ...



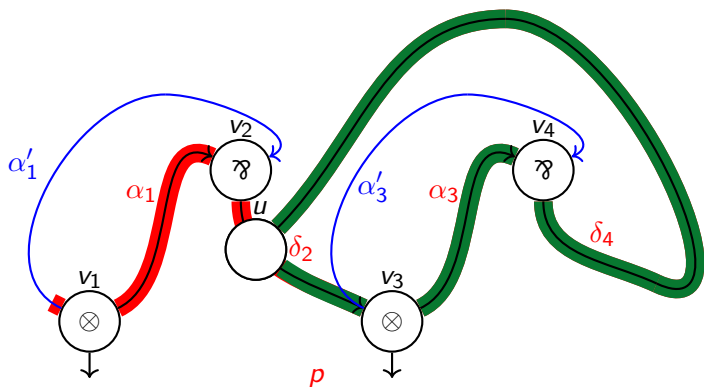
Plan

- 1 We build a **switching path** p by visiting vertices, halting our construction only on a vertex corresponding to a last rule. ✓
 - 2 We prove p **cannot "loop"**: its vertices are distinct.
- We get a contradiction by looking at the first vertex u appearing twice

No Going Back on a Descending Path



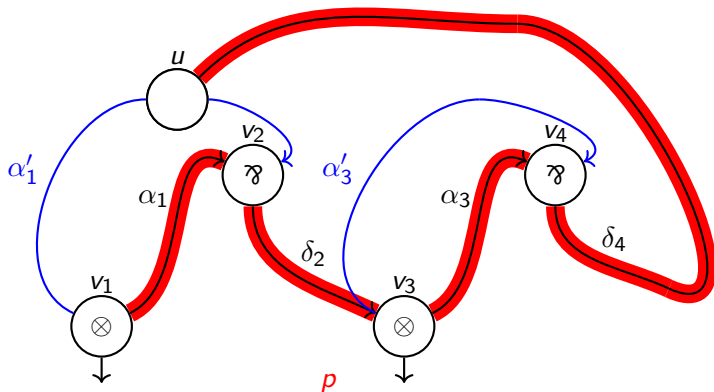
No Going Back on a Descending Path



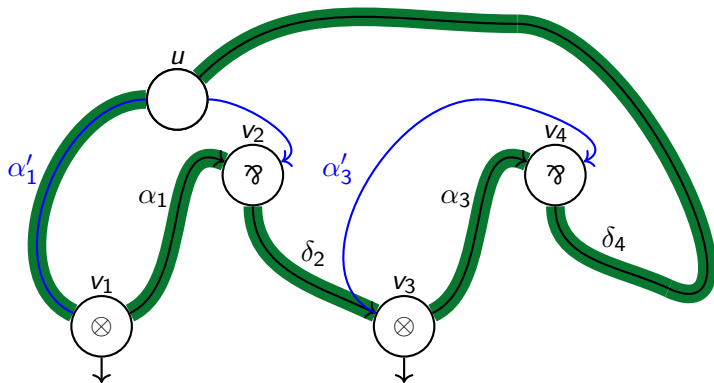
This is a switching cycle!

(no  at u as we took a downward edge in δ_2)

No Going Back on an Ascending Path



No Going Back on an Ascending Path

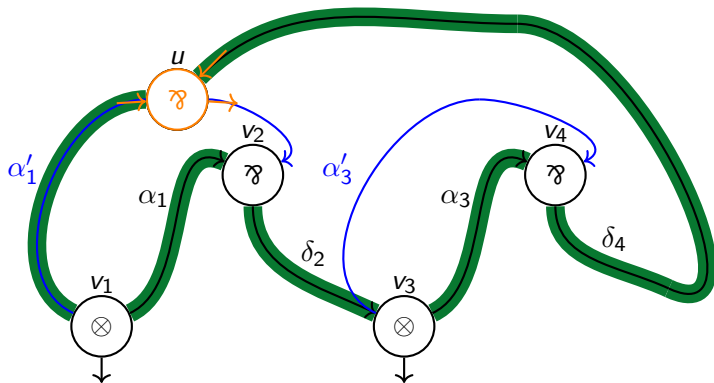


Counting: this has at least as many  as $\alpha_1 \cdot \alpha'_1!$

But we lost one at v_2 , so:

- there is none between u and v_2
- the green cycle takes both premises of u which is a $\wp$

No Going Back on an Ascending Path

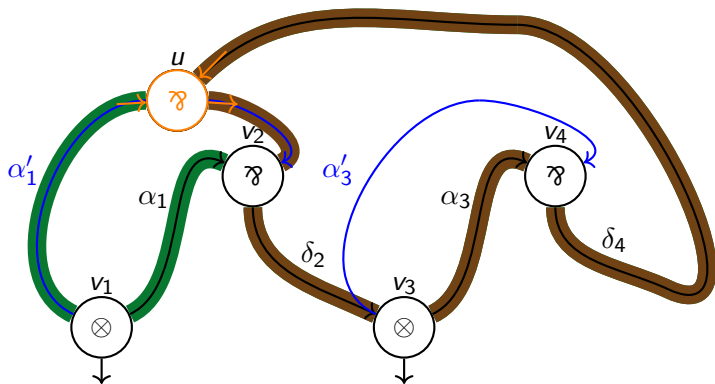


Counting: this has at least as many ∞ as $\alpha_1 \cdot \alpha'_1!$

But we lost one at v_2 , so:

- there is none between u and v_2
- the green cycle takes both premises of u which is a ∞

No Going Back on an Ascending Path



Counting: this has at least as many  as $\alpha_1 \cdot \alpha'_1!$

But we lost one at v_2 , so:

- there is none between u and v_2
- the green cycle takes both premises of u which is a $\mathfrak{I}$

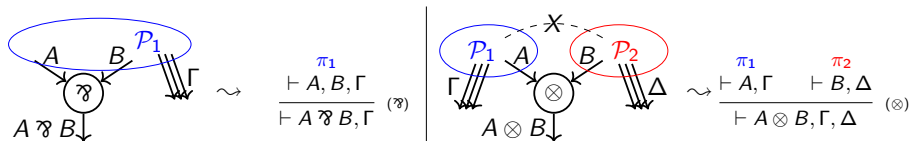
This is a switching cycle!

Summing-up

Plan

- 1 We build a **switching path** p by visiting vertices, halting our construction only on a vertex corresponding to a last rule. ✓
- 2 We prove p **cannot “loop”**: its vertices are distinct. ✓

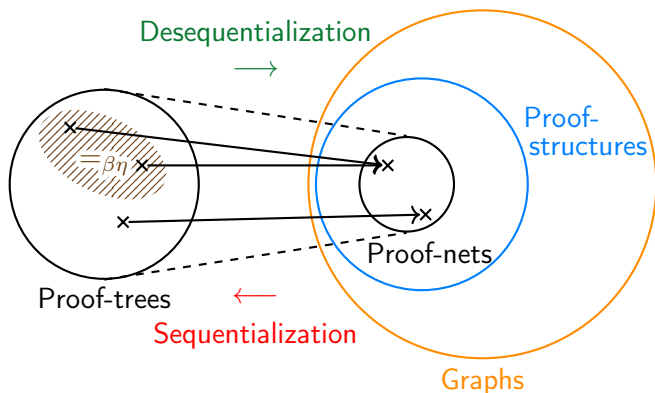
→ we get a $\otimes$ or $\wp$ corresponding to a last rule, remove it and conclude by induction hypothesis



- This solves the case when the proof-net has a $\wp$ or a $\otimes$
- If it has only ax , easy: by correctness it has exactly one, this is the image of an ax -rule

Concluding Remarks

- We only used acyclicity, not connectivity
→ this proof extends to **MELL proof-nets** (with/without *mix*)!
- Proof formalized in 
- Strong links with more standard **graph theory**: Kotzig's theorem on perfect matchings, Yeo's theorem on edge-colored graphs, ...
- The **quotient** of proof-trees by proof-nets is well-known



References



Link to these slides:



 **Handbook of Linear Logic (draft)**, <https://ll-handbook.frama.io/ll-handbook/ll-handbook-public.pdf>

<https://ll-handbook.frama.io/ll-handbook/ll-handbook-public.pdf>

↪ [this proof written with a different presentation](#)

 **Yeo's Theorem for Locally Colored Graphs: the Path to Sequentialization in Linear Logic**, Di Guardia, Laurent, Tortora de Falco, Vaux Auclair, <https://doi.org/10.4230/LIPIcs.FSCD.2025.16>

↪ [this proof written in graph theory](#)

 **Yeo's Theorem for Locally Colored Graphs: the Path to Sequentialization in Linear Logic (unpublished)**, Di Guardia, Laurent, Tortora de Falco, Vaux Auclair, <https://arxiv.org/abs/2603.03262>

↪ [extension to MALL proof-nets](#)